

AJAY GARG

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SUMMARY

Final-year B.Tech CSE (Cloud Computing) student at SRMIST with hands-on experience designing, building, and operating a 4-service microservices architecture deployed on AWS EC2 with Docker, Nginx, and GitHub Actions CI/CD. Strong foundations in Java and Python with a focus on DSA, OOP, REST API design, and backend development. NPTEL Elite certified in NLP (IIT Kharagpur) and co-author on research into autonomous self-healing cloud infrastructure using Prometheus/Grafana observability and automated remediation.

EDUCATION

B.Tech, Computer Science & Engineering – Cloud Computing

SRM Institute of Science and Technology (SRMIST)

2023 – 2027

- CGPA: 8.58/10

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

DSA: Arrays, Trees, Graphs, Hash Maps, Dynamic Programming, Topological Sort, BFS/DFS, DSU

OOP & Design: Object-Oriented Design, SOLID Principles, Design Patterns, REST API Design, Microservices

Cloud & AWS: AWS EC2, Docker, Docker Compose, Nginx, GitHub Actions (CI/CD), Prometheus, Grafana

Backend & DB: FastAPI, Node.js, Express.js, JWT/OAuth 2.0, PostgreSQL, MySQL, MongoDB

Frontend: React.js, TypeScript, Tailwind CSS, Responsive Design

AI / ML: LangChain, NLP, LLM Integration, FAISS, Vector Embeddings, GenAI Tools

PROJECTS

VEDA — Agentic AI Platform (Sole Developer) Jan 2025 – Present

FastAPI · Docker · Nginx · AWS EC2 · Prometheus/Grafana · FAISS

- Ownership: owned the complete SDLC — system design, OOP-based service decomposition, implementation, AWS deployment, CI/CD pipeline, and ongoing operational monitoring — as the sole engineer on the project.
- Distributed systems: designed 4 independent FastAPI microservices (Orchestrator, Planner, Memory, Executor) behind an Nginx load balancer on AWS EC2; services communicate via internal REST APIs with per-user JWT auth middleware.
- Operational excellence: configured Prometheus metrics scraping across all 4 services with Grafana dashboards for real-time health monitoring; GitHub Actions CI/CD pipeline cut deployment time from roughly [X min, manual] to [Y min, automated].

Digital Vault — Scalable Secure SaaS Platform Mar 2024 – Aug 2024

FastAPI · React/TypeScript · PostgreSQL · JWT · Agora SDK · Railway · Vercel

- Scalability & fault tolerance: designed stateless JWT auth (15-min access + 7-day single-use SHA-256-hashed refresh tokens) enabling horizontal scaling without shared session state.
- Production operations: deployed across 2 cloud providers (Railway + Vercel) with cross-origin auth, TOTP-based 2FA, account lockout after 5 failed attempts, and full audit logging.
- 12+ REST API endpoints (CRUD) across file storage, project management, notes, appointments; real-time P2P video via Agora SDK; responsive Tailwind CSS UI across mobile/desktop.

Email AI Agent — Distributed Automation Service Sep 2024 – Nov 2024

FastAPI · LangChain · Gmail API · OAuth 2.0 · NLP · PyTest

- Built LLM-powered agent (LangChain) that autonomously categorises email across 5 labels and generates contextual replies; OAuth 2.0 delegated access to Gmail API (3 scopes).
- Fault-tolerant design: 3-tier exponential retry logic handles LLM rate-limit failures gracefully; PyTest unit tests validate categorisation pipeline and OAuth token refresh.

Self-Healing Cloud Infrastructure — Research Project 2025

Python · Prometheus · Grafana · Docker · Autoencoders · AWS Cloud · ML

- Co-authored research on autonomous cloud remediation: Autoencoder anomaly detection on system metrics, Prometheus/Grafana observability, Docker-automated container restart.

ACHIEVEMENTS

Code Wizards 2.0 Hackathon — Full-Stack Engineer SRMIST · 2025

- Built both the React frontend and the FastAPI backend integration with a Qdrant vector database for an AI Compliance Management System, in a 4-person team — delivered a fully working demo within a 24-hour sprint.

CERTIFICATIONS

NPTEL Elite — Natural Language Processing

IIT Kharagpur / SWAYAM · 4 Credits

Jan – Apr 2026

Additional: HackerRank — Java Basic · Python Basic